

Parallel and Distributed Systems: **P**aradigms and **M**odels (**SPM**)

Academic year 2025-2026

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Outline

- Course objectives
- Lessons
- Timetable
- Exam
- Resources



**Parallel and
Distributed Systems**
Paradigms and Models (**SPM**)

Course objectives

- Understand parallel computing concepts and fundamentals
 - Explore various parallel systems architectures (shared-memory, distributed-memory, and GPGPUs)
 - Learn performance metrics and fundamental laws
 - Master the structured parallel programming approach
- Write efficient parallel code in modern C++ for a single multicore machine and for a compute cluster of multicore machines
 - C++ concurrency library, CUDA, OpenMP, FastFlow, MPI

Syllabus: three main pillars

Parallel Computing Concepts

- Performance metrics
- General Laws
- Models of Computations
- The Structured Parallel programming model

Shared Memory Arch.

- SIMD programming with Intrinsics
- SIMT programming in CUDA
- Concurrent Programming in modern C++, OpenMP and FastFlow
- Introduction to Lock-free programming

Distributed Memory Arch.

- Interconnection Networks
- Communication overhead, and Overlapping
- Synch/Asyn Communications and Collectives
- Distributed Programming with MPI
- Cluster Resource Management with SLURM

Course Path

- Bottom-Up approach
 - Writing efficient sequential code
 - Developing efficient parallel code on a multicore
 - Scaling to cluster-based systems
- High-level parallel programming: **the structured parallel programming approach**
- Throughout the course we will study standard and research tools for parallel programming

Lessons and question time



- Lessons will be recorded (no streaming) and uploaded to the MS Teams course portal
 - MS Teams code: **b7tb7di**
 - **We will use Teams to communicate and for last-minute updates**
- 72h course, 6h per week, Polo Fibonacci, building B
 - **Regular attendance is strongly encouraged**
- Days (please check the calendar at this [link](#)):
 - **Tuesday** 9-11 AM Fib C (building B)
 - **Wednesday** 11 AM -1 PM Fib C1 (building B)
 - **Friday** 11 AM – 1 PM Fib D3 ([building D](#)) <https://di.unipi.it/en/classrooms-and-buildings-of-the-polo-fibonacci/>
- Q&A:
 - Email me (massimo.torquati@unipi.it) with the subject: «SPM question time» or alike

Feb 16 – May 22, 2026
(+1 extra week to catch up, if needed)

Exam

- **It is mandatory to register on the esami.unipi.it portal to take part in any exam session**
- **Project + Oral Exam**
- Two project Options:

Option 1, Modular Project (4 modules)

- Assigned: First weeks of the course
- Submit: Progressive deliverables (code + report)
- Deadline: First exam session (all modules)
- Valid until: September 2026 session
- Best for: Students following lectures regularly

Option 2, Single Project

- Assigned: Last weeks of the course
- Submit: One deliverable (code + report)
- Valid for: All exam sessions of the academic year
- Can be your own topic (discuss with teacher)
- Best for: Students with specific project interests or those who prefer flexible scheduling

Exam: Project + Oral Exam

- **Option 1:** Modular project assigned during the first weeks of the course
 - It is structured into progressive modules (4 modules). Each module will have a soft deadline (suggested by the teacher)
 - You must submit four separate deliverables (source code + report for each module) according to the suggested schedule, and no later than the first exam session. Submitting all modules together only immediately before the first exam session is not acceptable for Option 1.
 - This option is intended for students who want to follow the course progression and develop the project in parallel with lectures
 - The discussion of the project and the oral exam must take place in the exam sessions up to the September session (included)
- **Option 2:** Announced during the last weeks of the course
 - A different project will be assigned at the end of the course
 - This project is valid for all exam sessions of the academic year
 - Could also be a topic of your choice (to be discussed with the teacher beforehand)
 - The deliverable is one report describing the project and the source code

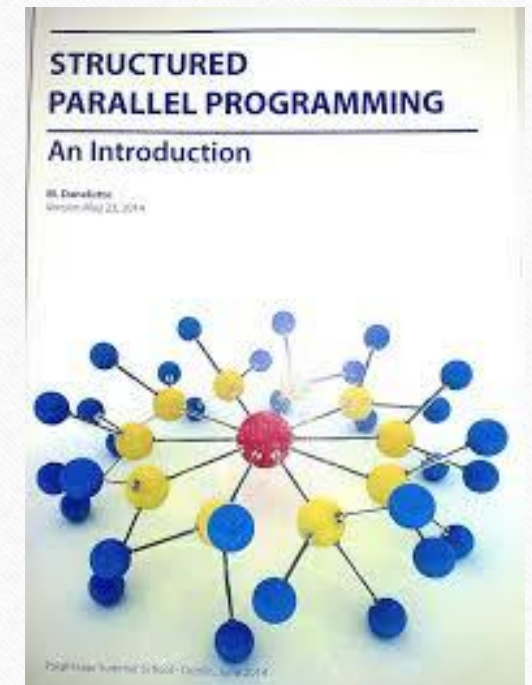
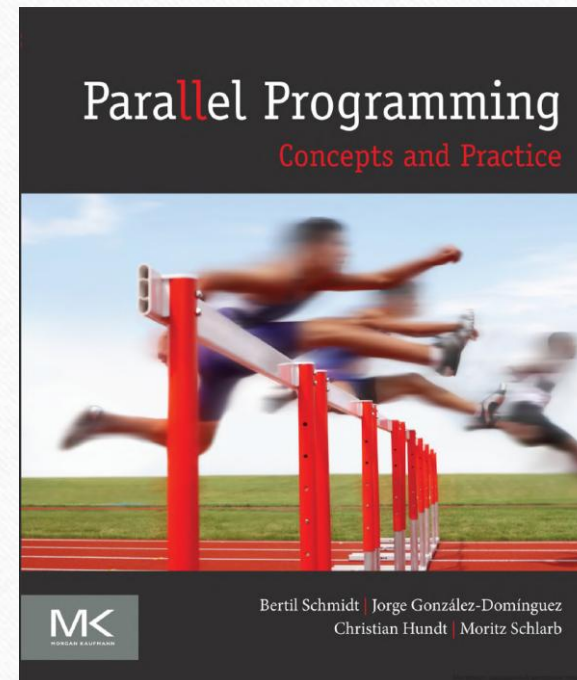
Exam: Project + Oral Exam

- Format:
 - Project demonstration (live on spmcluster.unipi.it → bring your laptop), discussion on the source code and the implementation techniques you have adopted, justify the implementation choices you made, justify why your implementation scales or does not scale, evaluate alternative solutions
 - Theory questions covering all course topics
- Important notes:
 - **Option 1:** Exam must be taken by September 2026
 - **Option 2:** Valid for all academic year sessions
 - Project discussion + theory part in the same session

Course Material

- Book: «Parallel Programming: Concepts and Practice» by S. Bertil et al.
- Professor Danelutto's course notes: «Structured Parallel Programming»
- Slides from lectures and additional notes
- Scientific papers and source code examples

Everything except the book will be provided by the teacher.



HW resources

- Compute cluster:
 - spmcluster.unipi.it (IP address 131.114.51.114) front-end node (where you have to login)
 - 8+1 internal nodes 10-Gbit Ethernet (where you have to compile and execute your code)
 - The +1 node is different from the others and equipped with an NVIDIA A30 GPU
 - Based on the Rocky Linux 9.3 OpenHPC
- To have access to the course machines, **you have to send me by email your public SSH key**
 - A brief document is available in the course portal with the info to gain access to the course machines

Assumed knowledge

- Basics concepts of computer architectures
- Knowledge of Operating Systems (scheduling, synchronization mechanisms, ...)
- Basics concepts of concurrent programming (threads, processes, mutexes,)
- **Proficiency in the C** (heap/stack allocation, pointers, passing arguments by value/reference, recursion, function pointers, etc...)
 - If you have gaps in C programming, it is paramount to work on filling them incrementally
 - Our primary programming language will be modern C++